

From Euclidean Space to Curved Surfaces

1. Euclidean Space and the Metric

In R^n , the Euclidean distance between two points $\vec{x}, \vec{y} \in R^n$ is given by:

$$d(\vec{x}, \vec{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}$$

This arises from the **inner product** (dot product):

$$\vec{x} \cdot \vec{y} = \sum_{i=1}^n x_i y_i$$

From this, we get length:

$$\|\vec{x}\| = \sqrt{\vec{x} \cdot \vec{x}}$$

and angle:

$$\cos \theta = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|}$$

2. Example: Unit Vector Sum in R^3

Let $\vec{v} = \mathbf{i} + \mathbf{j} + \mathbf{k} = (1, 1, 1)$, then:

$$\|\vec{v}\| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

3. Curved Geometry: Surface of a Sphere

In curved spaces (e.g. the surface of a sphere), the Euclidean metric no longer applies globally. Instead, the shortest path between two points lies along a **geodesic** — the spherical equivalent of a straight line.

Teaching Analogy: London to Paris Flight

- In flat geometry, shortest path = straight line through the Earth. - On Earth's surface, shortest path = arc of a great circle. - This introduces **intrinsic curvature**: locally flat, globally curved.

Geodesic distance $\neq$ Euclidean distance